Software Requirements Specification (SRS)

(Extreme Calendar)

Authors:

Instructor: Tianzhong Ding

Preface:

The expected readership of this document is anyone using the Extreme Calendar app. The reason for the creation of this app is to provide a free, easy-to-use calendar that has many different features for users who can then utilize as many or as few of them as possible. As of right now, this will be version 1.

1. Introduction: Kyle, Mostafa

The Software Requirements Specifications (SRS) is designed to document and describe the agreement between the customer and the developer regarding the specifications of the proposed software product, Extreme Calendar. Its primary purpose is to provide a clear and detailed statement of user and system requirements that will serve as a reference for the design, development, testing, and validation of the application. This document establishes a common understanding of what the system is expected to accomplish and the scope within which the development will occur.

This SRS is organized into multiple sections that will logically separate the requirements into structured and easily referenced components. Within this document, both user requirements and system requirements are described in detail, outlining the functionalities that the system must

provide and how users will interact with those functions. The specification also clarifies the assumptions, limitations, and operational environment of the Extreme Calendar application.

Furthermore, this document aims to define the system's functionality, external interface requirements, performance attributes, and design constraints that will guide implementation. The terminology and descriptions used throughout this specification are intended to remain clear, consistent and unambiguous to ensure accurate communication among stakeholders/clients and to support effective development and maintenance of the software system.

1.1 Purpose- Mostafa

Defining and describing the functions and specifications of the Extreme Calendar application is the primary goal of this software requirements specification. This document clearly outlines the systems intended uses, key features, and required functionality as determined by the project scope.

The intended audience of this document is our primary Extreme Calendar system customer. the CS3 instructor Mr.Ding, the spring 2026 semester CS3 group 5 members as well as the other students attending CS3 that will require access to such documentation.

1.2 Scope : Kyle, Mostafa

The Web Application will allow the user to keep up with their daily goals and monitor task completion. Data will be saved locally in a csv file for the user, such as their account-Username and Password. (Extreme Calendar will require users to create an account secured by a password before accessing the application’s features. After successful authentication, users will be

presented with a dashboard displaying the current day, date, and time, along with upcoming appointments and an option to set a financial goal. The application will provide a variety of interactive functions, including save, import, export, view summary, world clock, budget, edit, search events, filter, settings, format, and logout). The Calendar will allow the user to create events within each day window of a month , delete those same events, search for events created, personalize the user's time-zone(military time maybe), allow the user to designate events to Work/Family/Personal calendars, and personalize more in calendar settings. The user will be allowed to see the calendar in 4 different viewpoints; day, week, month, and year point of views.

Extreme Calendar will operate under several constraints. The system will not support calendar event sharing or collaboration features. It will not integrate with voice assistants such as siri, even on macOS systems. The application will not include printing capabilities for calendar events and will offer only limited formatting customization. Additionally the system will not support bluetooth synchronization or external divide syncing. These constraints are intended to keep the application lightweight and focused on local personal productivity management.

2. Overall Description: Mostafa

This section includes details about what is and is not expected of the Extreme Calendar system, including functionalities that are internationally excluded and assumptions that will guide the development of the application.

2.1 Product Perspective

The Extreme Calendar will be a downloadable application that will contain numerous functions for organizing one's tasks. It is designed for a user who needs many different organization features for their life and wants to organize all their tasks into one place. It will be available to anyone who is able to run Java on their device and should not require the internet except for the weather feature.

User Requirements

Evan

2.2 Product Functions(Functional Requirements) :

This app will provide a number of functions that are listed below;

1. The Calendar app shall determine the current date and time(based on the user's local time)
    a. The ability to display time in different cities - > allow the user to submit which cities they want to see. (The user's device will get the time from their city from their own desktop device).
    b. The ability to tell you the current weather and forecast
2. The app shall display a calendar in weekly, monthly, and yearly format, and allow the user to view any week, month or year they desire.

a. User-created and automatically generated events shall be tracked on the calendar and have editable dates and times.

3. The app should be customizable
    a. The ability to set a theme (background)
    b. Aspects of it can be resized
    c. It shall have a settings menu
        i. The settings will be nested in a sidebar, which will minimize when not in use
        ii. Settings should allow the user access to all the submenus / toggled functions.
4. It shall give the user a summary of their past week on Sunday.
    a. It shall also give the user a summary of their upcoming week
5. The app shall support the creation and editing of various calendar events
    a. Users shall be able to create and edit events
        i. When creating events they must specify a work, personal or school event.
        ii. Allow the user to search for events and specify the type of event when they create it
        iii. Allow the user to have categorized events and uncategorized to allow personalization
        iv. Users shall be able to create recurring events
        v. Undo/redo functionality
    b. Events shall be able to be automatically generated and then edited by the user

c. The ability to have the same events tracked with multiple calendar systems e.g. Julian Calendar, Islamic Calendar.

d. The app shall organize these events

i. The app shall have a search bar for events

ii. The app will be able to filter event categories.

1. Sub Calendars for different categories

iii. Allow customer to search for events

e. Set alerts / reminders

i. Ability to snooze

ii. Birthday/Anniversary options

6. Allows users to log in and out of the system

a. Different save data for each user

7. The user will have the ability to add a maximum spending budget for the week

a. The user can input how much they have spent and can see what percent of their budget they have used up for the week.

8. The ability to export the calendar

a. Export CSV -> (*allowing the user to export in a range of dates)*

b. Export printable calendar

9. The app will save info into a csv file

a. Shall have an auto-save feature

10. Add a progress bar that can be used for different tracking goals. You can adjust all of the settings manually but there could be some questions about it.

a. Users shall be able to track fitness information e.g. calories burned from time spent working out.

Nonfunctional (Define Observable Characteristics or Quality of System, etc) :

11. The program shall not be laggy.
    a. The UI shall be responsive to user input
12. It shall have a simple and easy to understand UI
13. Availability of Product - usage to students and all around adults(When will the app go down for Maintenance purposes?)
14. Reliability: user-input must be sanitized and limited to known safe inputs to prevent corruption of data or unexpected program behavior.
    a. Any input saved to a CSV file must not contain characters used to separate fields
    b. When the user is creating their event-details a character limit will be imposed

15. This app will rely on libraries and internal method calls before considering API integration, which requires an internet connection.
16. The program shall have error handling for both developer and user purposes
    a. During development logging and debug messages shall be more extensive in order to facilitate testing, extraneous debug features shall be removed prior to publishing in order to improve performance.
    b. The program shall record logs of its operation which can be referred back to in the case of unexpected behaviour or a crash.
    c. Error messages for the following :

i. Save/Load errors

ii. Failed API requests

iii. Improper user input

17. The UI should look cool and have a distinct aesthetic to differentiate the app.
18. The program shall be modular and extensible so that more features may be added in future.
19. The program shall be maintainable with good documentation.

2.3 User Characteristics

The typical Extreme Calendar user is anyone able to run Java applications on their devices. It is required that the user be familiar with using simple applications with their keyboard and mouse. The user is also likely to be someone who desires a more comprehensive calendar program with better aesthetics than typical.

The user will be allowed to schedule upcoming events involving school/work, personal, or family obligations. (NonFunctional)The user will be showcased a screen that allows them to “Login-in" or to “Create-Account”.

If the user clicks “Create Account” ; the user will be brought to a screen that will allow them to create a username and password. Next, they will be brought to the next screen that will display our three Calendar sections of -> Work , Personal, or Family. Whichever section that is chosen by the user; will become their designated calendar to create events. They will be allowed to switch from the three Calendar sections to differentiate the events that are occurring in their life.

If the user clicks “Log-In” ; the user will be asked for their username and their password. If they cannot remember their password (**We Might want a built-in/out-of- the- box functionality for ‘account linking’ if possible or we can just leave it and not make this part too complicated)**

Constraints - **> What the Calendar App should not do, Languages that we are forced to use, what will possibly take us more time and resources to reach the desired completed date for sections of the Calendar app**

2.4 Constraints - Christien

Security will not be a constraint and the user and password information can be stored in plain text as the app will be running locally, so anyone with direct access to the device would be trusted. No sensitive data will be stored as it is only storing budgets, events, and tasks. Security In order to run the software a person will have to be able to run a Java application on their device and should be familiar with using basic applications. Our development team is also working under the constraint of time as we are developing everything in around 10 weeks. The weather function will only be available when the device running the application is connected to the internet as the API requires that and either the user must enter a location or have location services enabled.

(More Constraints involving the SpringBootAPI will be put here)

2.5 Assumptions and Dependencies

Assumptions:

- After the user creates their account info involving Username and Password, they will be unable to change them afterwards
- The user will not be able to implement plug-ins
- The user will be able to access Calendar with or without Internet
- Users will be using up-to-date-computers, while attempting to access the Calendar App
- Application will only be tested on Windows 11

Dependencies(External and Integrated Factors of the Calendar App will rely on**:**

External -

- Operating systems that will be utilized -> Spring BootAPI
- Third Party Libraries for - Calendar rendering(Name libraries here), Date/Time Handling(here), Notifications services(here) , and UI framework
- We will be using this API https://open-meteo.com/ for receiving the data for a fourteen day weather forecast.

Integrated -

- Accessibility standards will be handled through our Username/Password system

Overall, some capabilities that could produce Scope Creep during this applications development will be the organization of libraries that be used for the Calendar App, the integration of notifications being possible for usage, and the time constraints of the presentation of this Application

(Moon Phases)

(Questionnaire about the user's

2.6 Divide and Allocation of Requirements(Might Not Need this Section)

3. Glossary - Christian

API (Application Programming Interface): A mechanism that allows two different software systems to communicate and share data with the other.

Button: A place within the GUI that the user can click to interact with to have an action occur

Calendar: A GUI showing the day, month, year and will also show tasks.

CSV File: A file that will hold the data for the different events that you have created and added to the calendar.

Event: A class that will have subclasses of school, work, personal life and all will have to have a date and description at least.

Functional Requirements: The functions that the application must perform in order to meet the needs of the users.

Non-Functional Requirements: The constraints on the system as a whole, as well as developmental and time constraints.

School Task: A school object that is a subclass of event. It will have a date, description and class in the constructor.

Personal Task: A personal object that will be a subclass of event. It will have a date and description at least in the constructor.

Work Task: A work object that will be a subclass of event. It will have a date and description at the least.

Use Case Diagram: A diagram showing the possible uses and actions the user can take within the application.

User: The operator of the software.

User Requirements: The requirements that are necessary to accommodate the user's needs and wants for this application.

(These sections will be filled in the future weeks)

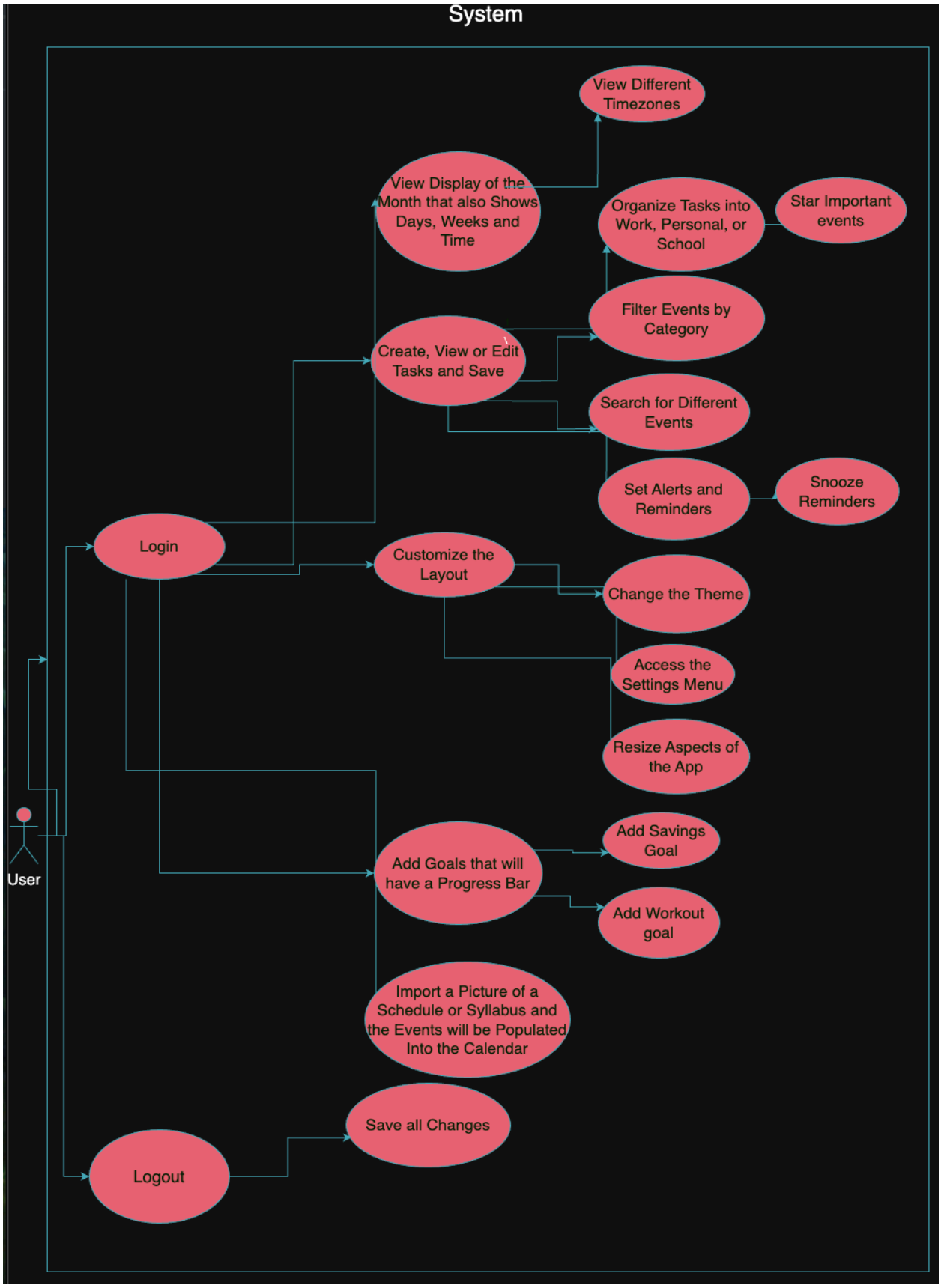
System Architecture

System Requirements(Details Specification of System behavior)

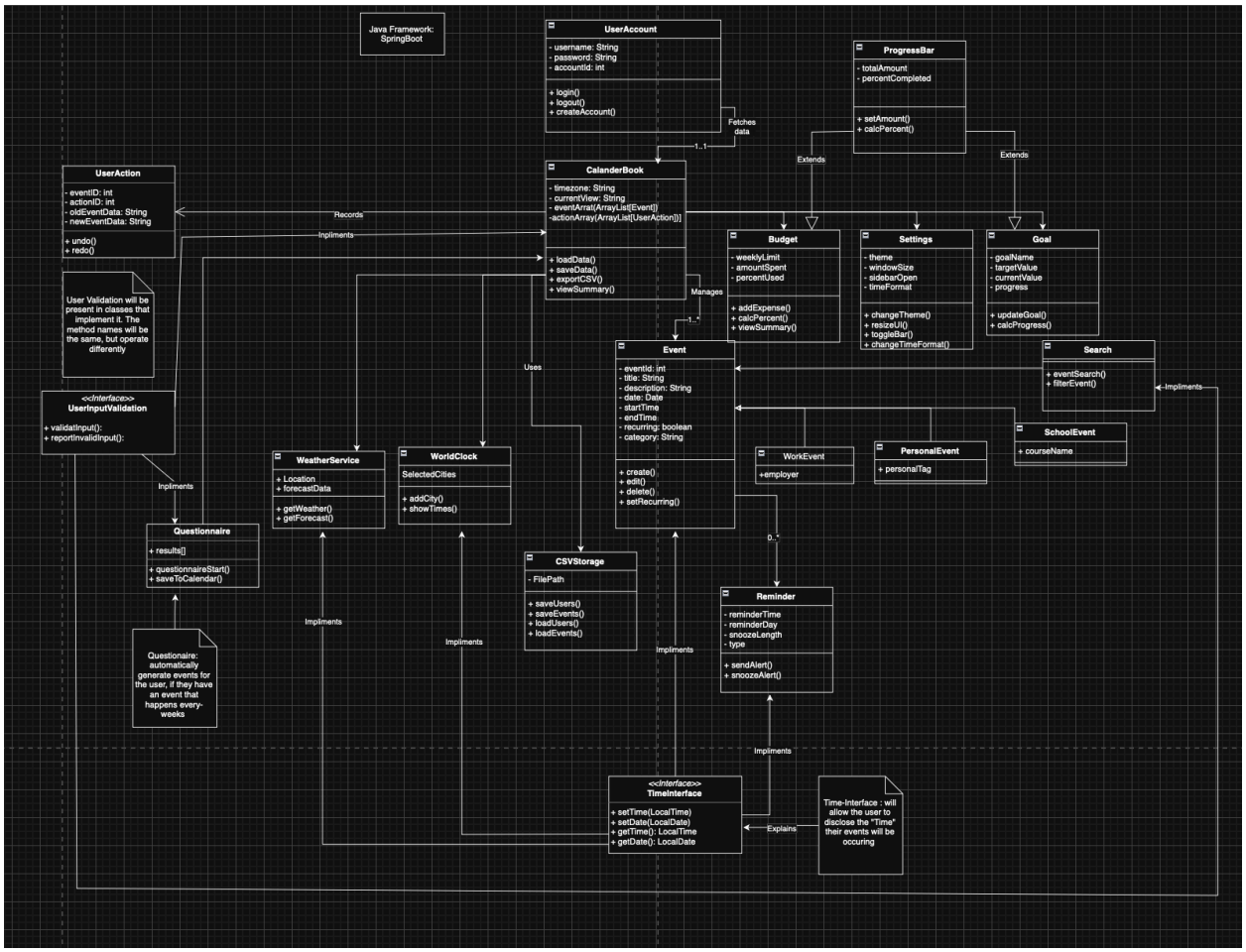
1. If the user tries to switch calendar sections without confirming an event to be added to the calendar that they are on(Work, Personal, Family) ; they will receive a **Warning**! That notifies them that " The event info that you are currently creating has not been confirmed. Proceeding with 'Calendar Section switch to (**whichever calendar section they tried to switch to)** will result in the event being deleted"
2. If the Developers agree to make this a Web Application, privacy policies, safety policies, and security policies would need to be reinforced or implemented into the Product

3.

System Models

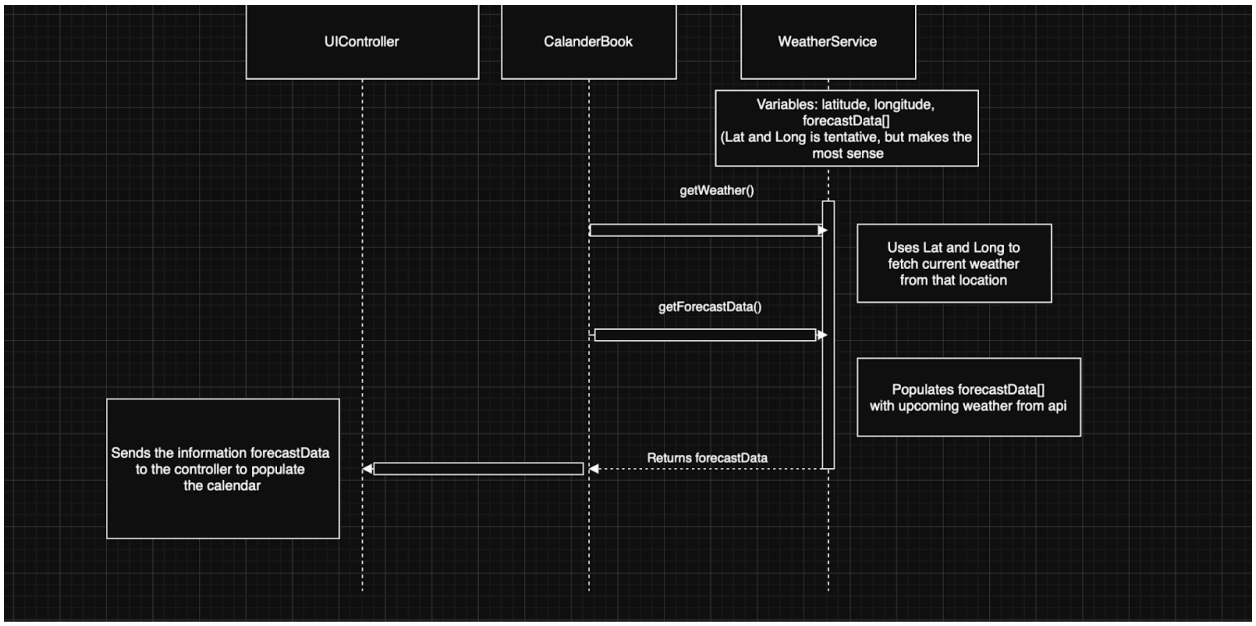

System
View Different Timezones
View Display of the Month that also Shows Days, Weeks and Time
Organize Tasks into Work, Personal, or School
Star Important events
Filter Events by Category
Create, View or Edit Tasks and Save
Search for Different Events
Set Alerts and Reminders
Snooze Reminders
Login
Customize the Layout
Change the Theme
Access the Settings Menu
Resize Aspects of the App
User
Add Savings Goal
Add Goals that will have a Progress Bar
Add Workout goal
Import a Picture of a Schedule or Syllabus and the Events will be Populated Into the Calendar
Save all Changes
Logout

Class Diagram



WeatherService Class

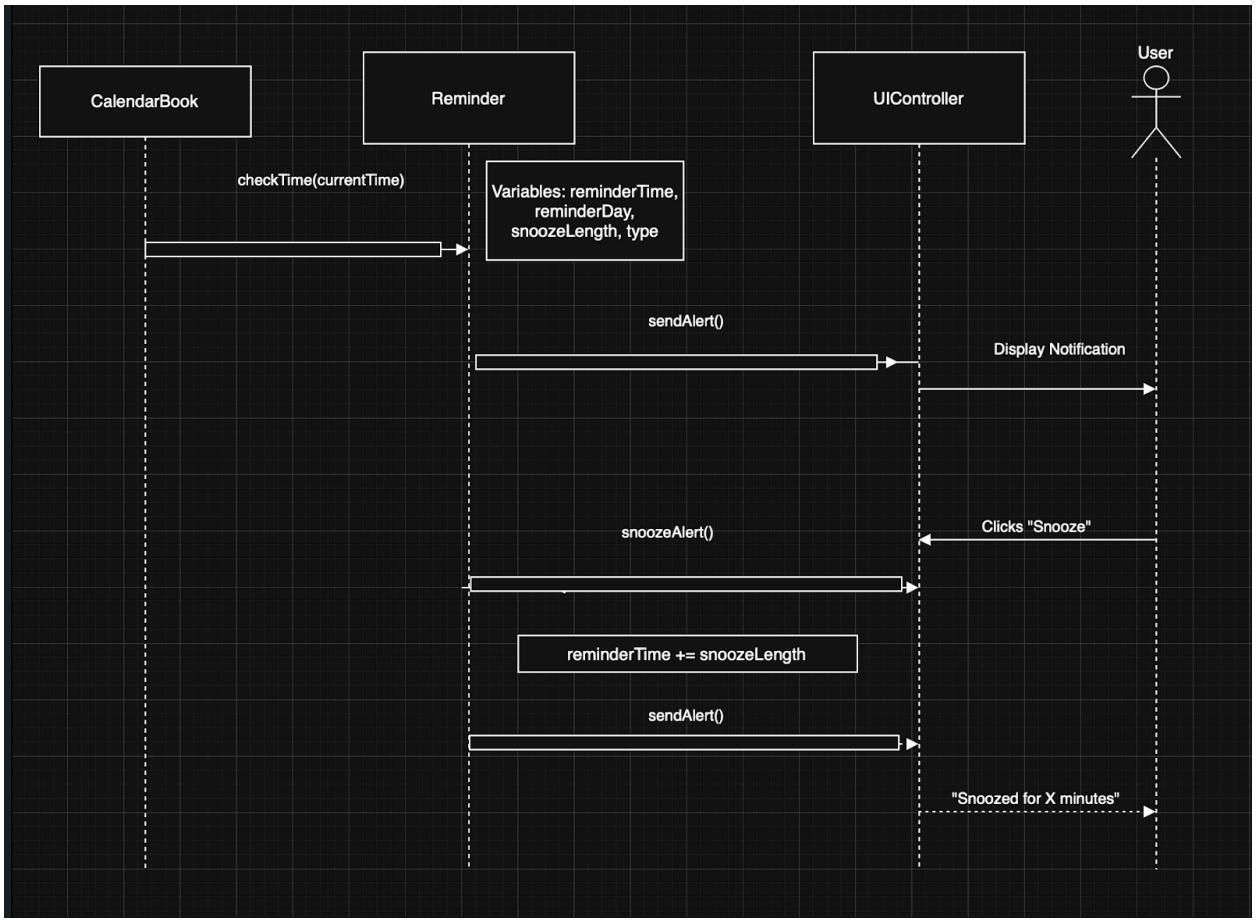
The WeatherService class will handle the API calls for getting the weather data. It will hold the location data and the forecast data. There will be a getWeather() function to get the current weather. There will also be a getForecast() function which will get the data for the weather forecast for the next seven days.



Reminder Class

The Reminder class will handle sending notifications such as reminders of upcoming events and will also be able to be snoozed or be reoccurring. The reminderTime variable will hold the schedule time for the reminder and the reminderDay will hold the date. The type variable will hold what kind of event it is and the snoozeLength will hold how long before the

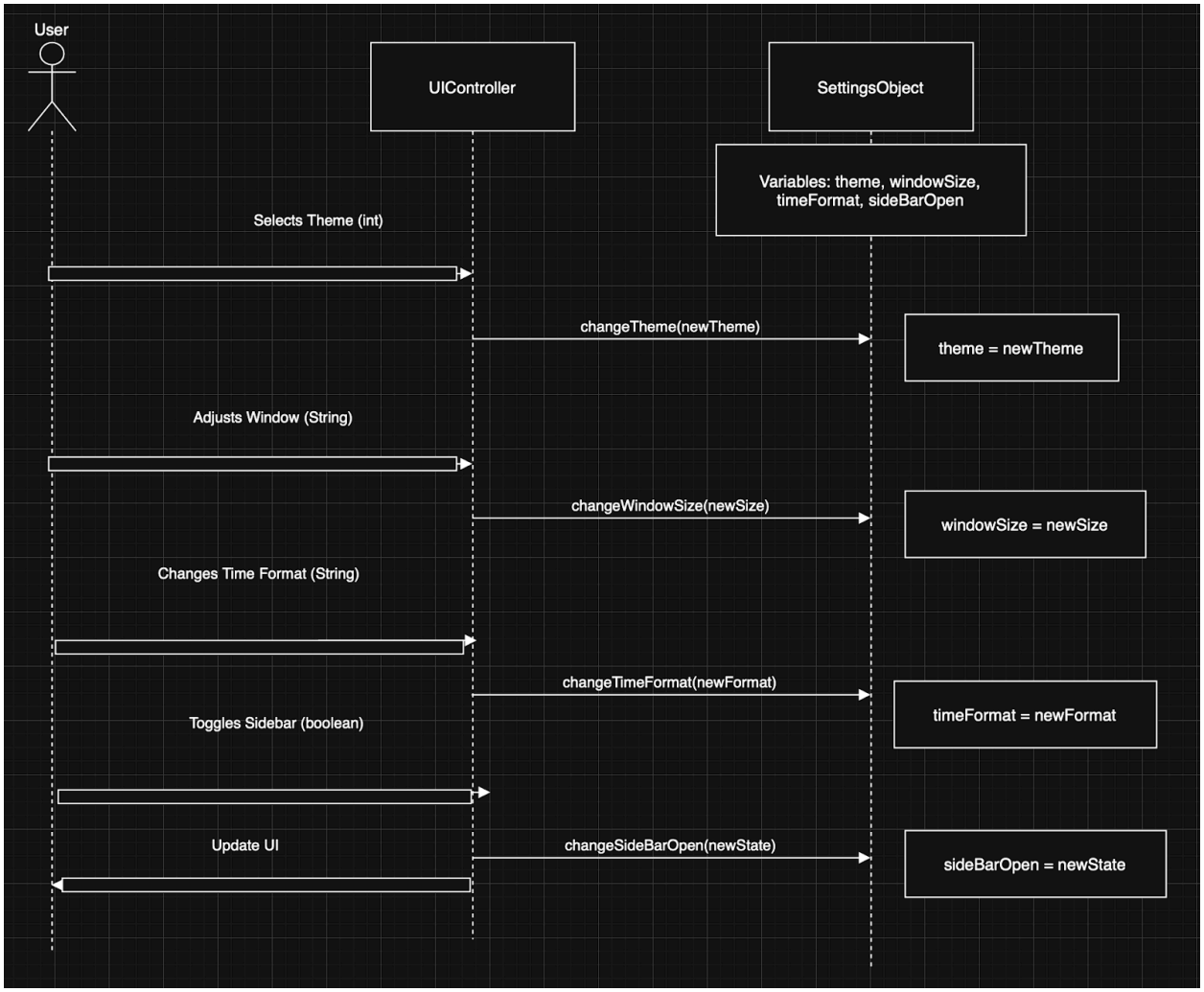
reminder sends another alert. The sendAlert() function will send a notification to remind the user about the event. The snooze() function will silence it and snooze.



Settings Class

The settings class will handle much of the customizability of the calendar. It will hold the theme, windowSize, sideBarOpen, and timeFormat variables. The theme will be the color of the calendar and the windowSize will be how large the application is on the user's screen. There will also be a variable timeFormat holding the current time format. The changeTheme() function will be used to change the theme of the application and the resizeUI() will be used to change the

window size. The toggleSideBar() function will open or close the sidebar acting like an on/off button. The changeTimeFormat() function will be used to change the time format between dd/mm/yy and mm/dd/yy formats.



CalendarBook Class

Stores Events, UserActions, and ProgressBars in program memory. Handles saving and loading events and progress bars. Allows for Events, ProgressBars, and undo/redo functionality to be accessed.

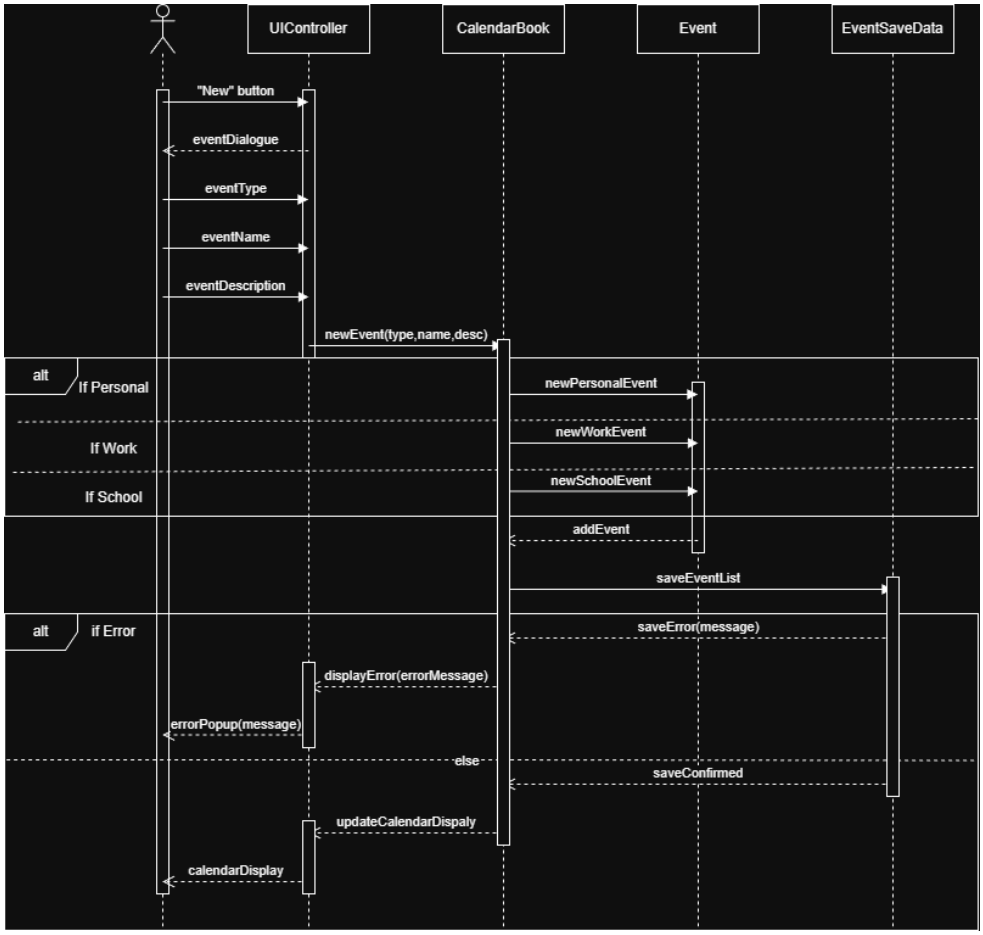
Event

Template for storing data for a calendar event. Generic superclass for more specific event types.

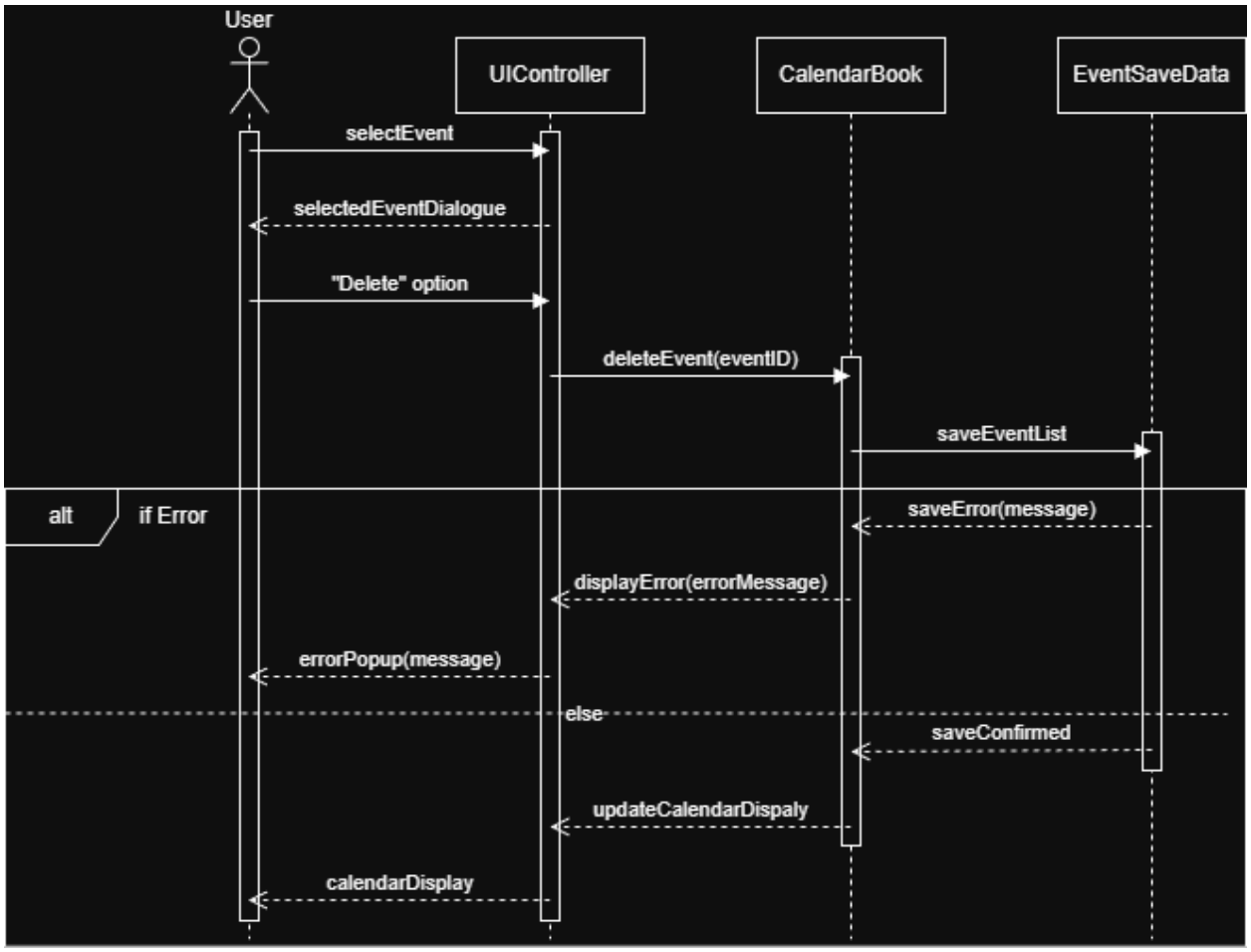
TimeTracking

Interface for handling date and time information for objects which either keep track of the current date/time or have a date/time associated with them.

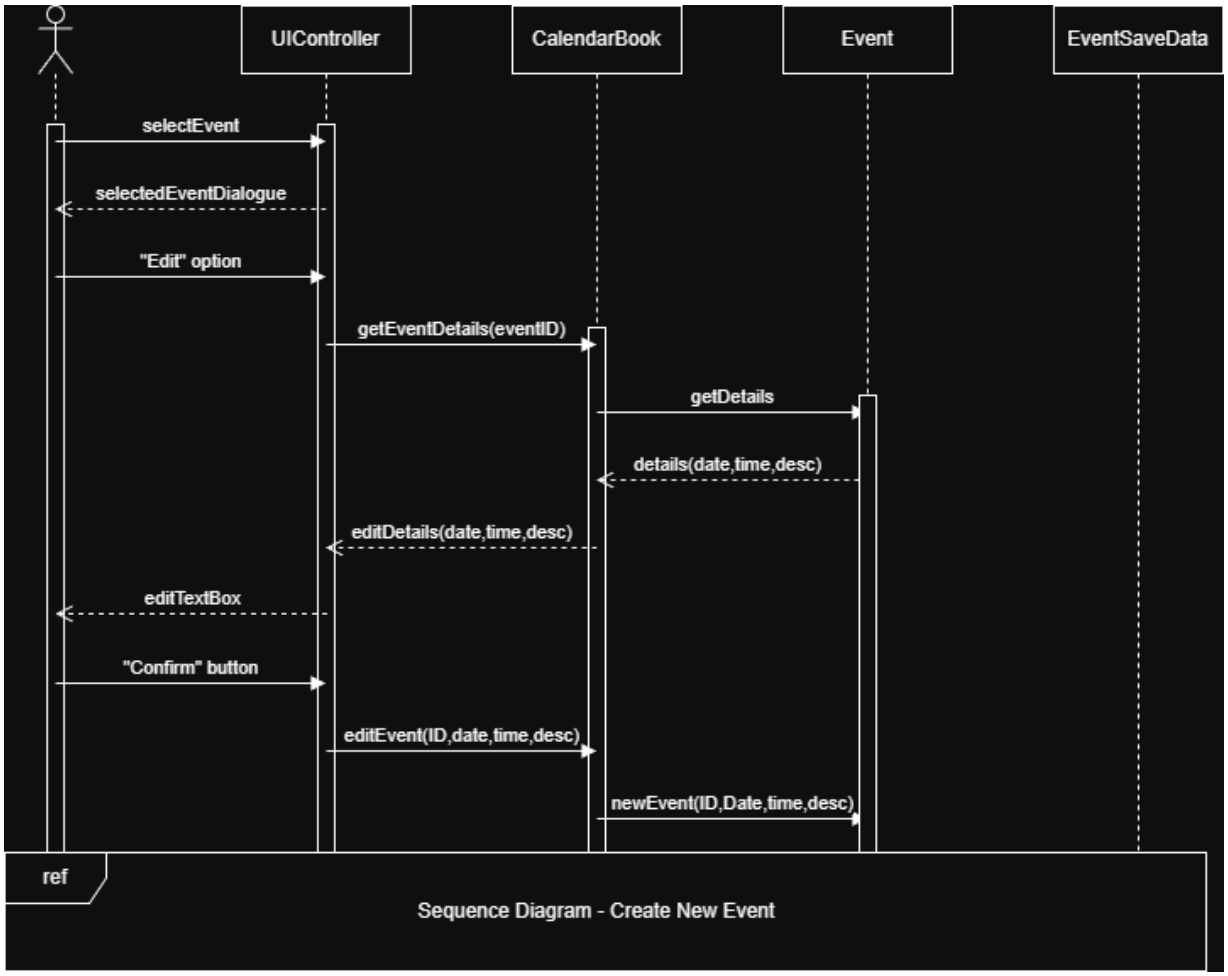
Create Event



Delete Event



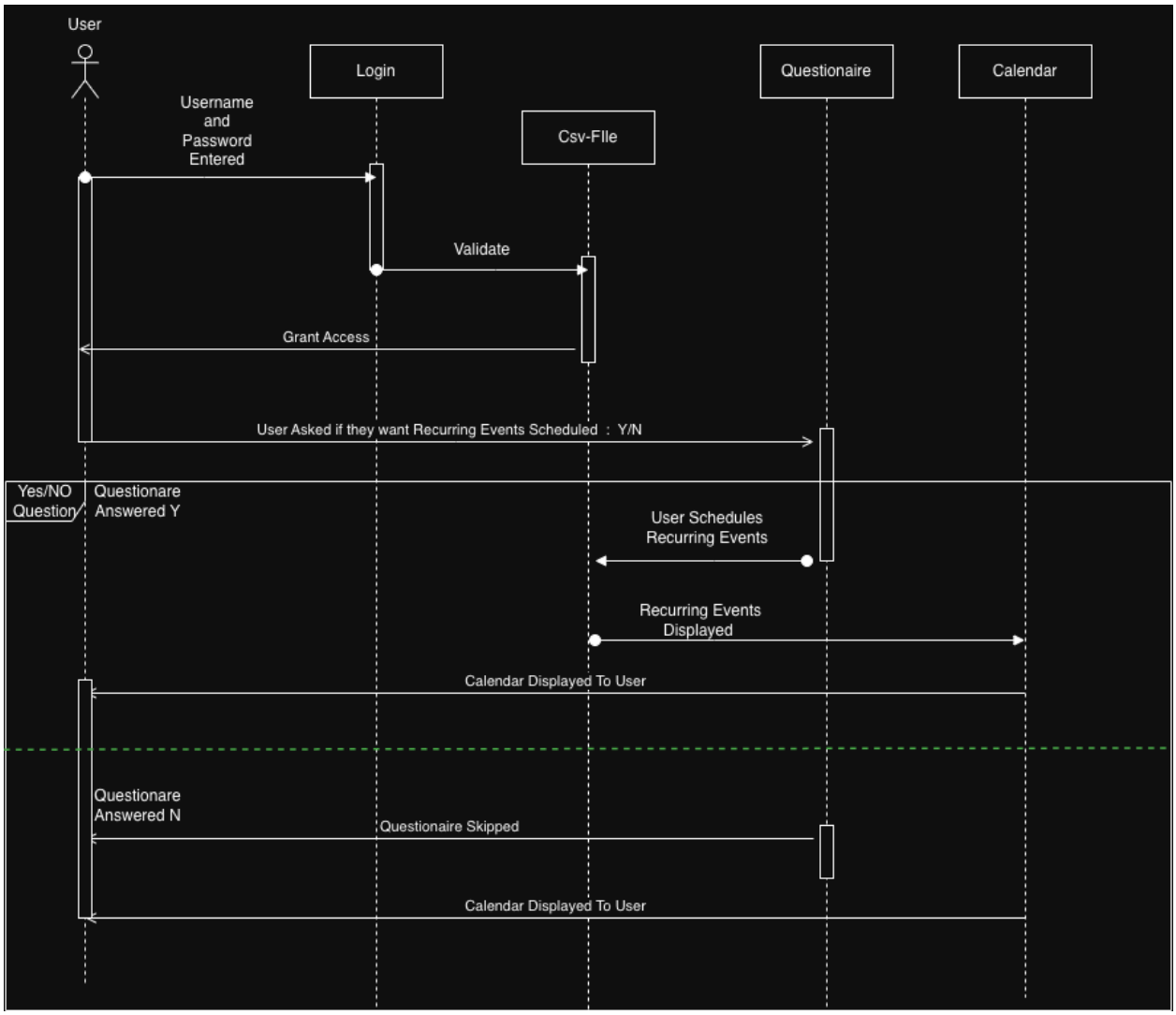
Edit Event



Questionnaire Class

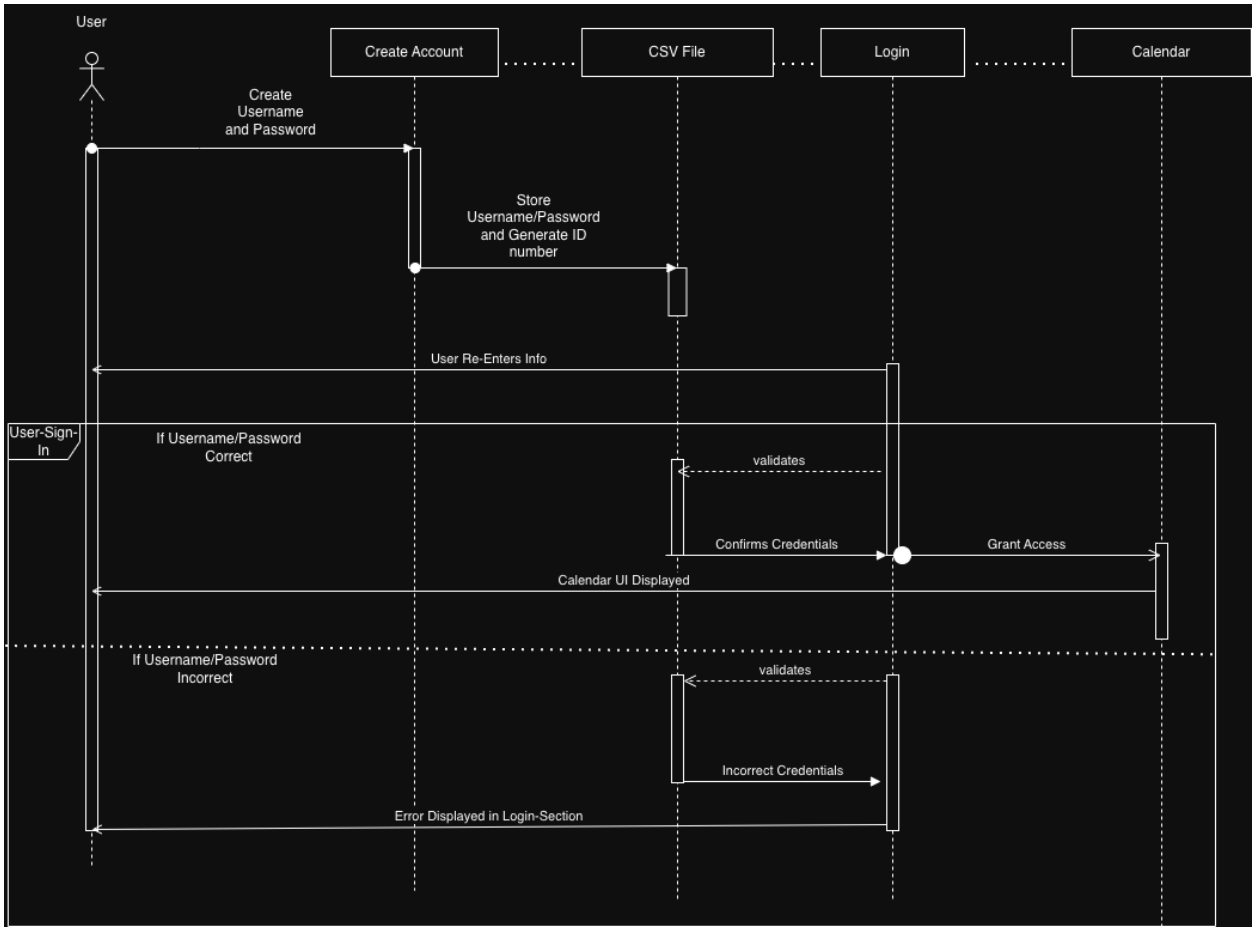
Serves as an easy method for the users to implement recurring events to their calendar, before accessing their CalendarBook. They will be prompted on the "Date" , and "Time" of the recurring event. Recurring events occur "each week, on the same day and time".
Recurring events can be stacked on the same day and time, but the limit will be two a day and if the user attempts to add more, a prompt will appear " Only Two Events A Day". The

questionnaire can be skipped, but adding "recurring events" will not be restricted. The option will show when the user attempts to create a new event within their CalendarBook.



User Class

Will hold users' "Username ,Passwords and UserIDs". When a user creates a username and password, it will be stored in a CSV- file and an ID number will be generated randomly for the user to link to their account and CalendarBook. If a user's username or password is incorrect, the login screen will display a prompt that says - "Username or Password is Incorrect. Try Again".



Index